TERMINAL VALUE MASTERY

A Deliberate Practice System under the Damodaran Framework

This workbook is designed to transition your theoretical understanding of Terminal Value (TV) into institutional-grade execution mechanics. In an intrinsic discounted cash flow (DCF) valuation, terminal value routinely accounts for 60% to 95% of total enterprise value. Consequently, structural or logical errors in its estimation invalidate the entire model. Follow this guide sequentially to build real, error-free valuation skills.

Beginner Exercises

Focus: Mechanical accuracy and understanding the algebraic levers of the Stable Growth Model.

Exercise 1: The Base Case Mechanics

Scenario: You are valuing a firm whose high-growth phase ends in Year 5.

- Expected Free Cash Flow to the Firm (FCFF) in Year 5 = \$100 million
- Cost of Capital ($WACC$) in perpetuity = 8%
- Stable growth rate (g_n) = 2%

Task: Calculate the Expected FCFF in Year 6 ($FCFF_6$) and the Terminal Value at the end of Year 5 (TV_5).

Exercise 2: The Reinvestment Linkage

Scenario: A firm reaches maturity in Year 10. The analyst projects Year 11 Operating Income ($EBIT(1-t)$) to be \$250 million. The firm's stable growth rate is 3%, and its mature Return on Invested Capital (ROIC) is expected to be 10%.

Formula Note: $Reinvestment\ Rate\ (RR) = g_n / ROIC$ and $FCFF = EBIT(1-t) \times (1 - RR)$

Task: Calculate the required stable Reinvestment Rate, the Year 11 FCFF, and the TV_{10} assuming a Cost of Capital of 9%.

Exercise 3: The Degrowing Cash Cow

Scenario: A legacy media company is entering its terminal phase. In Year 5, its FCFF is \$80 million. Due to structural industry decline, it is expected to shrink at a constant rate of -2% per year forever. The cost of capital drops to 6% as it pays off all debt.

Task: Calculate the TV_5. *(Pay close attention to the algebraic signs in the denominator: $WACC - g_n$).*

Intermediate Exercises

Focus: Making realistic judgment calls on inputs based on macroeconomic and lifecycle constraints.

Exercise 4: Enforcing the Growth Cap

Scenario: You are analyzing an Indian tech firm. The analyst team hands you a DCF where the stable growth rate (g_n) in Year 10 is set at 6.5% in USD terms. The current 10-year US Treasury rate (Risk-Free Rate) stands at 4.2%.

Task: Evaluate this input. Explain what is fundamentally wrong with a 6.5% perpetual growth rate in USD, and calculate the maximum dollar-denominated terminal value if Year 10 FCFF is \$500 million and $WACC$ is 9%.

Exercise 5: Changing of the Guard (Risk Transition)

Scenario: A high-growth EV startup currently has a Beta of 1.8, a Pre-tax Cost of Debt of 8%, and a Debt-to-Capital ratio of 15%. In Year 10, the company is expected to reach stable maturity.

- **Contextual Data:** Mature automotive peers have an average Beta of 1.05, a Pre-tax Cost of Debt of 5.5%, and a Debt-to-Capital ratio of 40%. The Risk-Free Rate is 4%, and the Equity Risk Premium (ERP) is 5%. The corporate tax rate is 25%.

Task: Calculate the high-growth $WACC$ and the mature $WACC$. Which one must you structurally use to discount the terminal value equation?

Exercise 6: The ROIC Convergence Call

Scenario: A consumer goods company has a Cost of Capital of 7.5% in its stable phase. It possesses a historically robust brand moat.

Task: You must choose between two terminal assumptions for Year 11 onwards:

1. The moat completely fades: $ROIC = WACC = 7.5\%$
2. The moat persists indefinitely: $ROIC = 12\%$

If Year 11 $EBIT(1\text{-}t)$ = \$150 million and g_n = 3%, calculate the TV_{10} under both scenarios. Quantify the percentage difference in valuation.

Advanced Exercises

Focus: Navigating structural shifts, structural asset termination, and conflicting market signals.

Exercise 7: The Reinvestment Mismatch Paradox

Scenario: You are reviewing a colleague's model for a renewable energy firm. In Year 5, Capital Expenditures are projected at \$300 million and Depreciation is \$300 million (Net CapEx = 0). The model seamlessly projects a perpetual growth rate of 3.5% from Year 6 onwards.

Task: Identify the structural flaw in this model. If the firm's stable ROIC is capped at 10%, correct the Year 6 cash flow assumptions to make the model internally consistent, assuming Year 5 ***EBIT(1-t)*** was \$400 million and ***WACC*** is 8%.

Exercise 8: The Terminal Value Horizon Shift

Scenario: You are valuing a biotech company with a blockbuster drug patent that expires exactly 15 years after your 5-year high-growth forecast period. After Year 20, generic competition forces cash flows to drop to zero.

Task: Instead of a perpetual growth model, apply Damodaran's alternative: a **growing annuity**. If Year 6 expected FCFF is \$120 million, ***WACC*** is 10%, and growth during the stable 15-year annuity phase is 2%, calculate the TV_5 using the growing annuity formula:

$$TV_n = [\ FCFF_{n+1} / (WACC - g_n)\ ] \times [\ 1 - (\ (1 + g_n) / (1 + WACC)\ )^t\ ]$$

Exercise 9: The Currency Inflation Trap

Scenario: You are valuing an Argentine subsidiary. The high-growth cash flows are modeled in Argentine Pesos (ARS) due to high local inflation (expected stable inflation = 20%). The risk-free rate in ARS is 22%. Your peer insists on capping the terminal growth rate at the US risk-free rate of 4%.

Task: Resolve the conflict. Explain how inflation dictates the growth cap across different currencies, and determine the mathematically correct growth cap for the local ARS valuation.

Real Company Task: ITC Limited

Note: ITC Limited represents an excellent real-world case study for terminal value due to its mature, cash-generative tobacco business transitioning cash flows into growth segments like FMCG and Hotels.

1. Where to Go

Open the latest Standalone Financial Statements section of the **ITC Limited Annual Report**. Locate the **Profit & Loss Statement** and the **Cash Flow Statement**.

2. What to Extract

- **Operating Income:** Extract Profit Before Tax (PBT) and Finance Costs to calculate ***EBIT = PBT + Finance Costs***.
- **Effective Tax Rate:** Calculate ***Current Tax / PBT***.

- **Reinvestment Inputs:** Extract Capital Expenditures (Purchase of Property, Plant, and Equipment) and Depreciation/Amortization from the Cash Flow Statement.

3. Step-by-Step Execution & Judgment Points

- **Step 1: Compute After-Tax Operating Income**
  Calculation: $EBIT \times (1 - t)$
  Judgment Point 1: Is the current effective tax rate sustainable in perpetuity? If India's statutory corporate tax rate is 25.17%, should you use ITC's actual historical rate or normalize it to the statutory rate for the stable phase?
- **Step 2: Establish the Growth Cap (g_n)**
  Judgment Point 2: Observe the current yield on the 10-year Indian Government Bond (G-Sec). If it is ~7%, this serves as your nominal local currency risk-free rate proxy. What stable growth rate will you assign to ITC? Damodaran notes that for a mature firm, it should not exceed this risk-free rate. Will you choose 4%, 5%, or 6%?
- **Step 3: Fix the Stable ROIC and Reinvestment Rate**
  Calculation: $\text{Stable Reinvestment Rate} = g_n / ROIC$
  Judgment Point 3: ITC has historically enjoyed an ROIC above 30% due to its tobacco monopoly. In perpetuity, as competitive dynamics shift and the company reinvests into lower-margin sectors, will you assume its ROIC drifts down to its cost of capital (~10-11%), or will you award it a perpetual competitive advantage (e.g., ROIC of 15% or 20%)?
- **Step 4: Calculate Terminal Value**
  Calculation: Apply the standard model using your adjusted Year n+1 FCFF based on the chosen reinvestment rate and a mature $WACC$.

Modeling & Estimation Drills

1. Spreadsheet Architecture (From Scratch)

Build a dynamic excel layout from scratch incorporating the following rules:

- **Global Inputs:** Risk-free rate, ERP, Stable Beta, High-growth Beta.
- **Forecast Rows:** Year 1 to Year 10 rows for EBIT, Tax Rate, Reinvestment Rate, and FCFF.
- **The Transition Check:** Create a conditional formatting formula: If Cell g_n > Cell `Risk-free Rate`, trigger a RED warning flag to isolate an illegal macroeconomic assumption.

2. Sensitivity Analysis Drill

Create a two-dimensional Data Table in your sheet evaluating TV_{10} across varying ranges:

- **Vertical Axis (Rows):** Cost of Capital ($WACC$) ranging from 7.0% to 11.0% in steps of 0.5%.
- **Horizontal Axis (Columns):** Stable Growth Rate (g_n) ranging from 1.0% to 4.0% in steps of 0.5%.

3. Scenario Analysis Narrative Template

Input Variable	Base Case	Bull Case (Moat Persists)	Bear Case (Disruption)
Stable Growth (g_n)	2.5%	4.0%	-1.0%
Terminal ROIC	9.0% (=*WACC*)	15.0%	6.0%
Stable Beta	1.00	0.90	1.20

4. Estimation Drill: The Missing Reinvestment Rate

Problem: You are valuing a private logistics firm. You have chosen a terminal growth rate of 3% and calculated a mature ***WACC*** of 9.5%. However, because it is private, you cannot reliably calculate its historical ROIC.

Damodaran Approach: Assume that for an average company in maturity, its Return on Invested Capital (ROIC) will naturally converge to equal its Cost of Capital (***WACC***) as competitive forces compete away excess returns over time.

Task: Apply this principle to estimate the required terminal Reinvestment Rate and calculate the ***TV*** if terminal ***EBIT(1-t)*** = \$50 million.

Mistake Analysis: The Bermuda Triangle Framework

1. The Growth Rate Illusion

Cause: Analysts map short-term or aggregate country GDP growth rates (e.g., 7% in India) directly into perpetuity equations.

Appearance: g_n = 7% while the long-term risk-free bond yield stands at 6%.

Avoidance: Strictly enforce the ceiling: $g_n \leq \text{Risk-Free Rate}$. The risk-free rate is the absolute nominal economic boundary for stable perpetual growth.

2. The Capital Expenditure Disconnect

Cause: Artificially setting CapEx equal to Depreciation in the terminal year to maximize free cash flow while leaving a positive growth rate in place.

Appearance: Net CapEx = 0, while g_n = 3%.

Avoidance: Never hardcode cash flows independently of growth. Derive terminal reinvestment dynamically from growth and ROIC choices ($RR = g / ROIC$).

3. The Big Startup Beta Trap

Cause: Forgetting to normalize the discount rate as the company scales down from high growth to steady state maturity.

Appearance: A company growing at a mature 2% in Year 11, but its cash flows are still discounted using a high-growth 15% cost of capital.

Avoidance: Implement a linear glide path that automatically transitions Beta and Cost of Capital to mature industry averages by the terminal year.

Mini Valuation Project: "Neo-Crafts Ltd."

> **COMPANY PROFILE & HIGH GROWTH INPUTS**
>
> Neo-Crafts is an artisanal e-commerce brand experiencing an initial high-growth surge, expected to hit steady-state maturity at the end of Year 5.
>
> - Year 5 Expected Operating Income ($EBIT$): \$20 million
> - Corporate Tax Rate: 30%
> - High-Growth Cost of Capital ($WACC$): 12%

Terminal Phase Parameters (Year 6+ onwards)

- **Risk-Free Rate:** 4% | **Equity Risk Premium (ERP):** 5.5%
- **Stable Industry Average Beta:** 1.0
- **Stable Pre-tax Cost of Debt:** 6% | **Stable Debt-to-Capital Ratio:** 30%
- **Stable Growth Rate (g_n):** Maximum allowed by Damodaran's principles.
- **Terminal Return on Invested Capital (ROIC):** Assumed to match the stable Cost of Capital.

Expected Deliverables

1. Calculate the stable, mature Cost of Capital ($WACC_{stable}$) incorporating the structural changes provided for Year 6+.
2. Determine the correct stable growth rate (g_n) and the corresponding stable Reinvestment Rate.
3. Calculate the explicit Expected Free Cash Flow to the Firm in Year 6 ($FCFF_6$).
4. Calculate the Terminal Value at Year 5 (TV_5).
5. **Quality Check:** What percentage of overall enterprise value does the Present Value of the Terminal Value represent if the Present Value of explicit cash flows (Years 1-5) totals exactly **\$42 million**?